

OPERA: Open-World Performance Evaluation for Biomedical Research Agents

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Abstract

We present OPERA (Open-World Performance Evaluation for Biomedical Research Agents), an evaluation framework for long-horizon, open-ended biomedical discovery tasks. Unlike closed-form benchmarks that score against fixed answers, OPERA captures the agent’s full reasoning trace—hypothesis evolution, evidence accumulation, and tool usage—and evaluates the *quality of scientific process* rather than final correctness. The framework introduces a three-persona pipeline (Task Proposer, Evaluator, Reviewer) with structured artifact handoffs, a rubric-driven LLM-as-judge scorer that uses two-pass scoring (evidence-only, then rationale-only) across multiple backends to surface disagreement rather than average it away, an independent falsification observer that derives a confidence estimate from the evidence trail alone, and a novel martingale score that quantifies whether hypothesis confidence updates are evidence-driven. We demonstrate OPERA empirically on a biomarker-discovery task using both Claude and Gemini agents, specify a second, structurally distinct discovery task to illustrate the framework’s generality, and argue that open-world process evaluation is necessary for benchmarking agents on tasks where ground truth is unknown.

1 Introduction

Large language model (LLM) agents are increasingly deployed on scientific discovery tasks: literature mining, hypothesis generation, experimental design, and data analysis [3, 9]. Yet existing benchmarks measure whether an agent reaches a known answer—a regime that fundamentally mismatches the structure of real research, where ground truth is unknown, tasks span weeks of effort, and quality is a property of the *reasoning process*, not just the conclusion.

We identify three specific gaps:

1. **Fixed-answer evaluation** cannot score open-ended tasks whose correct answer is not yet known to humanity [10].
2. **Single-turn or short-horizon tasks** do not capture the multi-step hypothesis revision that characterizes real discovery [10].
3. **Monolithic scoring** conflates tool-use breadth, evidence quality, and novelty—obscuring which agent capabilities matter.

OPERA (Open-World Performance Evaluation for Biomedical Research Agents) addresses all three. For (1), a task specification format encodes *evaluation criteria* rather than expected outputs, so a task can be scored without a known correct answer. For (2), a structured hypothesis ledger records the full trajectory of belief updates—not just the agent’s final hypothesis set—so multi-step revision itself becomes an object of evaluation, and a martingale score separately flags belief entrenchment: confidence that drifts independent of new evidence. For (3), a multi-stage reviewer pipeline disaggregates what monolithic scoring conflates: two-pass LLM judging—evidence-only, then rationale-only—scores evidence quality separately from the agent’s own rationale, multiple judge backends run in parallel so disagreement is surfaced rather than averaged away, and an independent falsification observer checks the agent’s expressed confidence against the evidence trail alone—all before an optional human expert review.

We use the SLE Biomarker Discovery task as a case study throughout this paper, run end-to-end from task specification through agent execution to LLM and human review. An agent receives a public single-cell RNA-seq cohort and must propose candidate cell-type-specific biomarkers with multi-modal supporting evidence—a question whose correct answer is not known until the candidates are validated in follow-up studies. Beyond this starting dataset, the agent has open-ended access to biomedical databases, literature, and computational tools, and may combine statistical analysis, literature search, database queries, and hypothesis revision however it chooses. §4 gives the full task specification and rubric.

2 Related Work

Open-world evaluation. Kapoor et al. [10] distinguish a spectrum of evaluation approaches from Q&A benchmarks and outcome-only agent benchmarks through to *open-world evaluations*: long-horizon tasks in real-world settings where success cannot be automatically graded and qualitative log analysis is necessary. They argue that as AI capabilities grow, evaluations must grow correspondingly complex, and that open-world evaluations provide early warning of frontier capabilities that benchmarks cannot surface. OPERA is an instance of this class, applied to biomedical discovery. Where Kapoor et al. focus on characterizing open-world evaluations as a methodology, OPERA contributes concrete infrastructure and metrics—a structured hypothesis ledger, process-level automated scores, and a rubric-driven multi-stage reviewer—that make open-world scientific evaluations reproducible and extensible.

Biomedical agent benchmarks. A cluster of recent benchmarks evaluates LLM agents on biological and biomedical analysis tasks, but all share a common assumption: a correct answer is known at evaluation time. BioMysteryBench [2] presents 99 professional-level bioinformatics problems paired with verifiable answer rubrics derived from data properties; BixBench [12] evaluates agents on 61 expert-curated computational biology capsules scored by binary LLM judgment; and BioML-Bench frames biological machine learning as a Kaggle-style competition with deterministic metrics against ground truth. AstaBench [4], the broadest of these, spans 2,400+ problems across the full research pipeline—literature understanding, code execution, data analysis, and end-to-end discovery—but its E2E evaluation still requires a defined outcome against which agent outputs are scored. More open-ended systems such as AutoDiscovery [1] and AI Scientist [11] automate hypothesis generation and testing but still rely on closed verification—a fixed dataset or executable experiment—to score outcomes. What none of these frameworks can measure is the quality of an agent’s reasoning *process* on a question whose answer is not yet known. When the task itself is a genuine open scientific question—what

biomarkers predict lupus flare in an understudied cohort?—there is no key to grade against. OPERA is designed precisely for this setting.

Belief update metrics and Bayesian trajectory scoring. A key insight motivating OPERA’s process metrics is that LLMs systematically fail to update beliefs correctly in individual sequential reasoning chains, even when their average behavior across independent queries approximates Bayesian rationality [5]. This failure manifests as belief entrenchment: whatever confidence an agent currently holds tends to be reinforced by subsequent reasoning steps, regardless of evidence. The Martingale Score [7] formalizes this as the Pearson correlation between b_t and Δb_t —a statistic that can be computed without ground truth and that has been shown to correlate with accuracy on domains where labels are available. OPERA adopts this metric directly, applying it to the confidence sequences recorded in the hypothesis ledger. For post-hoc hypothesis validation, OPERA draws on POPPER [8], which applies sequential falsification via e-value accumulation—rather than p-values, which require a pre-specified stopping time—to assess whether accumulated evidence clears a statistically rigorous bar. We adopt this framing for its interpretability as incrementally accumulating evidence, not for a proven guarantee: our implementation assigns e-values heuristically via an LLM (described below) rather than deriving them as likelihood ratios against a specified null, so the formal anytime-validity property does not carry over to OPERA’s reported confidence estimate. The practical implication of [5] for OPERA’s design is the requirement that the falsification observer be a separate model instance: asking the running agent to self-assess its evidence is unreliable precisely because the same model that generated the beliefs will entrench them.

LLM-as-judge scoring. Zheng et al. [14] establish that language models can reliably assess open-ended outputs against rubric criteria, forming the basis for our LLM-as-judge scoring stage. OPERA extends this in two directions: the two-pass protocol—evidence-only, then rationale-only—ensures scores reflect evidence quality rather than rhetorical fluency, and running multiple judge backends in parallel keeps disagreement visible rather than averaging it away. Criteria where backends diverge are flagged for the human reviewer once their own independent scoring is complete.

3 The OPERA Framework

OPERA structures each evaluation run as a pipeline of three well-separated roles—Task Proposer, Evaluator, and Reviewer—each with defined input and output contracts (Figure 1). Separation of concerns is a deliberate design choice: the agent under evaluation never has access to the rubric used to score it, and the scoring stage never communicates with the agent directly. This prevents the two most common failure modes in LLM-based evaluation: teaching to the test and post-hoc rationalization.

3.1 Task Specification and the Rubric Contract

A central design choice in OPERA is that evaluation criteria are specified independently of expected answers. Each task is defined by two artifacts produced by the Task Proposer: a machine-readable task specification encoding the scientific goal, termination conditions, and a structured rubric; and an agent-facing task prompt written in purely goal-oriented language that describes what a successful outcome looks like without prescribing methodology.

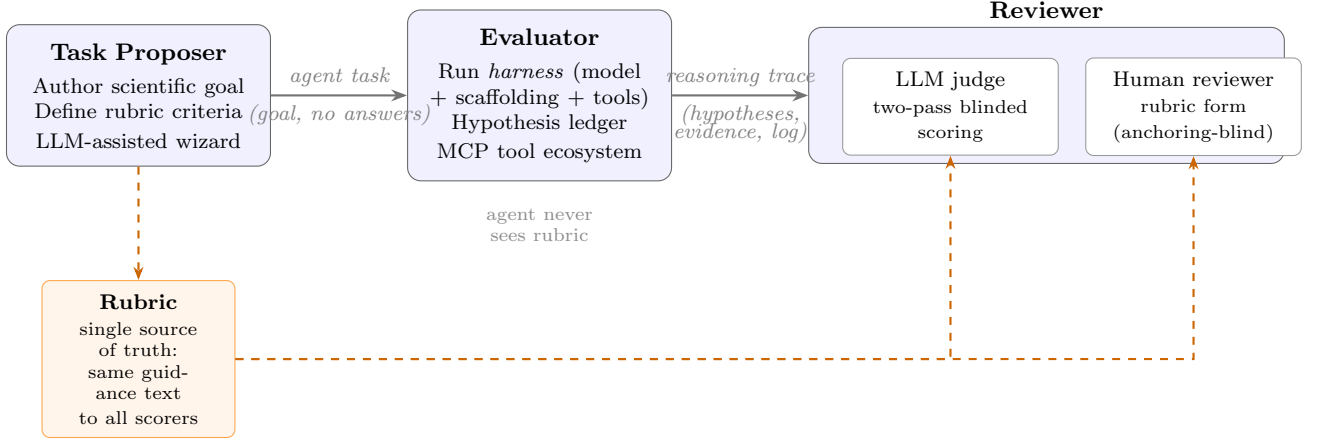


Figure 1: OPERA’s three-persona pipeline. Solid arrows carry run-time artifacts; dashed orange arrows show the *rubric*—authored once by the Task Proposer and injected verbatim into both the LLM judge and the human reviewer’s scoring form. This shared contract prevents criterion drift between automated and human evaluation. The agent under evaluation never has access to the rubric.

The rubric is the single source of truth for what constitutes quality in a run. Each rubric item carries a criterion identifier, a descriptive label, a scope (per-hypothesis or run-level), a numeric scale (typically 1–5, 1 weakest and 5 strongest, occasionally yes/no/partial), and a *guidance* string written for a domain scientist. Critically, the same guidance text is injected verbatim into LLM scoring prompts and rendered in the human reviewer’s interface—preventing criterion drift between automated and human evaluation and making the evaluation criteria fully auditable. An LLM-assisted authoring tool lowers the barrier to task creation by scaffolding both artifacts from a free-text description, though hand-authoring is equally supported.

3.2 Agent Execution and the Hypothesis Ledger

OPERA evaluates a *harness*—the full system comprising model, scaffolding, tool access, and any agent-specific prompting strategy—not an isolated model. Two runs using the same underlying model but different harnesses may produce meaningfully different belief trajectories, and OPERA’s metrics are designed to surface exactly that variation.

The Evaluator stage currently supports two widely-used agent harnesses—Claude Code and Gemini CLI—connected to a shared tool ecosystem via the Model Context Protocol (MCP). Because the Evaluator’s only requirement is that the agent record structured artifacts (hypotheses, evidence, run log) through defined interfaces, any MCP-compatible harness can participate without changes to the evaluation infrastructure. This cleanly separates harness-specific behavior from hypothesis tracking and scoring, which remain uniform across agent systems.

Agents operate with broad tool access by design: harness-native capabilities (shell execution, web search, file I/O) are unrestricted, and two MCP layers add structured instrumentation. The hypothesis-ledger MCP provides six operations for managing hypothesis state: recording a new hypothesis with an initial confidence score, updating confidence with supporting evidence and a rationale, and marking a hypothesis terminal (confirmed or refuted). A key constraint is the enforced separation of evidence from rationale—evidence entries must be factual (observed

measurements, database identifiers, statistical values, literature citations) while rationale entries are interpretive, explaining why confidence changed. This separation is a prerequisite for the two-pass scoring design described below. A second MCP layer provides access to over 600 pre-built biomedical tools spanning literature retrieval, genomics databases, pathway analysis, and drug-target interaction [6], enabling agents to pursue multi-modal evidence without bespoke integration work. The agent is free to use any combination of these capabilities; which tools it reaches for is itself a behavioral signal captured in the tool-diversity metric.

3.3 Multi-Stage Review and Scoring

The Reviewer stage operates entirely offline on the artifacts produced by the Evaluator and is designed to yield three complementary signals: automated process metrics, an independent LLM-based assessment, and optional human expert adjudication.

Automated metrics quantify the structural properties of the agent’s reasoning trajectory without any LLM calls. Four metrics characterize hypothesis testing behavior in context: the number of turns used (out of the allocated budget), the total number of hypotheses formed, the mean number of belief revision events per hypothesis (i.e., how many times the agent returned to update a hypothesis after new evidence), and the refutation rate (the fraction of hypotheses explicitly rejected). Reporting turns alongside hypothesis counts is essential for interpretation: an agent that formed five hypotheses in ten turns is meaningfully different from one that formed five hypotheses in 250 turns. Zero revisions on a hypothesis is itself a signal—it indicates the agent registered a candidate but never re-examined it in light of subsequent evidence. The same structural counting extends to individual hypotheses in Table 3, which reports each one’s own revision count alongside its evidence item count—the number of distinct entries logged in that hypothesis’s evidence list, typically one per citation or quantitative finding. Together these metrics characterize whether the agent engaged in genuine iterative reasoning or merely accumulated hypotheses without revisiting them.

Tool breadth is captured separately for each tool layer, using the normalized Shannon entropy $\hat{H}$:

$$\hat{H} = \frac{-\sum_i p_i \log_2 p_i}{\log_2 N}, \quad (1)$$

where p_i is the fraction of calls to tool i and N is the number of tools available in that layer. Normalizing by $\log_2 N$ yields a value in $[0, 1]$ regardless of layer size, making the two layers—the six hypothesis-management operations and the 600+ biomedical tools—directly comparable. A score near 1 indicates the agent explored the available tool space broadly; a low score indicates concentration on a small subset. Neither extreme is normatively better: in real research scenarios, an agent may legitimately rely on a few high-value tools repeatedly, so low $\hat{H}$ reflects research strategy as much as breadth.

A belief entrenchment indicator is computed as the Pearson correlation r between a hypothesis’s current confidence b_t and its subsequent one-step change $\Delta b_t = b_{t+1} - b_t$, pooled across all hypotheses in the run [7]:

$$r = \text{Pearson}(b_t, \Delta b_t). \quad (2)$$

A near-zero r indicates updates are unpredictable from the current confidence level, consistent with evidence-driven revision; a positive r signals entrenchment, where already-high confidence tends to keep rising regardless of new evidence.

The confidence values b_t are the agent’s *expressed* beliefs, reported through the hypothesis-ledger interface. They are not independently verified—an agent could in principle report arbitrary values. Two design choices bound this concern. First, the agent never has access to the

rubric, so there is no scoring target to game confidence values toward. Second, the falsification observer independently derives a confidence estimate from the evidence trail alone and compares it to the agent’s self-reported values (Eq. 3); systematic divergence between the two is a direct signal of overconfidence or anchoring. The martingale score therefore measures internal self-consistency of expressed beliefs, while the observer comparison measures their external grounding. In the pilot runs this grounding check pulled in opposite directions: the observer read more confidence than the agent claimed on every hypothesis in Run B (the agent conservative relative to the evidence), and less confidence than the agent claimed on every hypothesis in Run C (the agent overconfident relative to the evidence)—a contrast the martingale score alone, which only checks self-consistency, cannot surface.

Because the statistic is pooled across all $(b_t, \Delta b_t)$ pairs in the run, its reliability depends directly on the total number of revision events: runs with few revisions yield a highly variable estimate and should be treated as descriptive rather than conclusive. All automated metrics are computed offline from the recorded reasoning trace and included in the review report.

LLM-as-judge scoring uses a two-pass protocol designed to isolate evidence quality from agent rhetoric. In the first pass, the judge model receives only the factual evidence trail for each hypothesis—tool outputs, citations, quantitative findings—and scores each rubric criterion without access to the agent’s stated reasoning. In the second pass, the judge is shown the agent’s own rationale and asked to characterize agreement or divergence, but cannot revise the first-pass scores. This blinding ensures that a well-articulated but evidence-poor argument cannot inflate criterion scores. To reduce dependence on any single judge model, all configured backends run in parallel (in the pilot: Claude Haiku 4.5 and Gemini 2.5 Flash); when two or more backends score the same criterion and their scores disagree by two or more points on a five-point scale, the criterion is flagged as *contested*. This flag is revealed to the human reviewer after they submit their own independent ratings.

An additional falsification observer provides an external check on the agent’s expressed confidence. A separate LLM instance (Claude Haiku 4.5) reads only the accumulated evidence trail for each hypothesis—never the agent’s stated confidence or rationale—and independently estimates evidential support following the sequential falsification framework of POPPER [8], producing two outputs per hypothesis: a calibrated confidence estimate, and three candidate falsification criteria—concrete disconfirming findings that would undermine the claim if observed, proposed directly by the same evidence-only reasoning pass. The model assigns an individual e-value e_i to each evidence item from a calibrated heuristic scale (derived from stated p-values and effect sizes where available) and reports a confidence estimate C intended to approximate

$$C = \frac{E}{1 + E}, \quad E = \prod_i e_i, \quad (3)$$

where E is the product of the e-values it assigns to each evidence item. In practice, we find this self-reported C is consistently more conservative than literal multiplication of the model’s own e-values would imply—never less, by 0.05–0.27 across the pilot hypotheses we checked—suggesting the model discounts for evidence non-independence rather than applying the formula mechanically; it is this self-reported C , not a mechanically recomputed one, that we report throughout. Comparing C to the agent’s self-reported b_t provides a direct measure of overconfidence or anchoring. Crucially, the observer does not depend on whether the agent formally finalized a hypothesis in its report: a hypothesis the agent quietly accumulated strong evidence for, without ever raising its stated confidence or naming it a top candidate, is just as assessable as one the agent explicitly named in its final output. As a result, the falsification check runs against every non-rejected hypothesis in the ledger at run end, not just the ones the agent chose to

formally declare.

Finally, a review package is assembled for a domain scientist, presenting the task context, rubric, and full step-by-step agent log. All model-derived scores—both LLM judge scores and the falsification observer’s confidence estimate C —are concealed until after the human submits their own rubric ratings, preventing score anchoring. The falsification criteria are the one deliberate exception: shown to the reviewer before they score, since a falsification criterion is a framing aid rather than a score, and withholding it would serve no purpose the blinding design is meant to protect. This design allows independent human adjudication without requiring access to model APIs or evaluation infrastructure.

4 Pilot Tasks and Preliminary Results

We instantiate OPERA on two biomedical discovery tasks to demonstrate the framework’s generality: pilot runs and all automated/LLM-judge results below are from the first task; the second is specified (rubric and full instructions written and reviewable) but not yet run, and is reported here to show the task-specification format transfers to a structurally different discovery problem.

4.1 Task Instantiations

The first task, *SLE Biomarker Discovery*, asks an agent to identify novel candidate biomarkers for systemic lupus erythematosus (SLE) disease activity and flare prediction. The agent is provided a starting reference dataset—a 1.2M-cell single-cell RNA-seq PBMC cohort [13]—but is otherwise unconstrained: it may access any publicly available databases, literature, protein interaction networks, GWAS catalogs, drug-target repositories, and computational tools it can discover through the tool ecosystem. The rubric comprises seven criteria: four scored per hypothesis (novelty, evidence strength, experimental design quality, and clinical relevance) and three scored overall (reasoning quality, overall scientific quality, and whether the top hypotheses are worth pursuing in a wet lab); Appendix A gives the full specification.

The second task, *Protein Synergy Discovery*, asks an agent to investigate the synergistic (non-additive) effect of double mutations in a protein fitness landscape—e.g., the GB1 deep mutational scanning dataset—and determine whether that synergy is driven primarily by proximity to the protein’s active site or by physical distance between the two mutation sites. The agent must discover a suitable fitness-landscape dataset itself and justify its choice. The rubric comprises six criteria: two scored per hypothesis (novelty and evidence strength) and four scored overall (visualization correctness—e.g., never using a divergent color scale on raw, non-relative fitness values; interview quality—whether the agent asks clarifying questions about the data or biological context before analysis; dataset discovery; and overall scientific quality); Appendix B gives the full specification. We have not yet run agents on this task; we include its specification to illustrate that the task-definition format generalizes beyond a single reasoning regime—from a data-anchored biomarker identification task with an existing reference dataset to a structural-biology task where the agent must both find its own dataset and justify a mechanistic explanation for a quantitative pattern in it—and leave pilot runs and results on it to future work.

4.2 Results

We report automated and LLM-judge metrics from three pilot runs (30-turn budget) on the SLE Biomarker Discovery task, using two frontier model families under identical task specifications (Table 1). For open-ended discovery tasks there is no predetermined correct turn count; 30 turns is enough to form and revise hypotheses, but more compute allows deeper literature integration, tool exploration, and iterative refinement. Longer runs and human expert adjudication are ongoing; the results here establish that OPERA’s evaluation layers produce meaningful signal even at modest scale.

Table 1: Automated process metrics for pilot runs on the SLE Biomarker Discovery task. Runs A and B are two *independent* sessions of the same model (Claude Sonnet 4.6) on the identical task and prompt, illustrating genuine run-to-run variability rather than different configurations; Run C uses a different model family (Gemini 2.5 Pro). Turns: model invocations used; Hyp.: total hypotheses formed; Rev./Hyp.: mean revision events per hypothesis; Refuted: fraction explicitly rejected; Martingale r : belief entrenchment indicator (pooled Pearson correlation; n = total revision pairs).

Model	Run	Turns	Hyp.	Rev./Hyp.	Refuted	Martingale r
Claude Sonnet 4.6	A	31	5	0.4	20%	N/A ($n=2$) [†]
Claude Sonnet 4.6	B	31	3	2.3	0%	−0.43 ($n=7$)
Gemini 2.5 Pro	C	30 [‡]	3	1.0	0%	0.00 ($n=3$)

[†] $n < 3$ revision pairs; score undefined.

[‡] Budget-equivalent; Gemini CLI records no per-turn counter.

Automated metrics reveal qualitatively distinct search strategies. Across three runs at the same 30-turn budget, the agents produce strikingly different hypothesis trajectories (Table 3; Fig. 2 details Run B). Run A (Claude) spread across five hypotheses with shallow per-hypothesis investigation (0.4 revisions each) but explicitly refuted one candidate (20% refutation rate), indicating an active breadth-first falsification strategy—and used Bash-based scripting as its primary tool (69 shell invocations). Run B (Claude) converged on three candidates with deep iterative revision (2.3 revisions each) and no formal refutations—a concentrated, literature-driven style (WebSearch $\times 15$). Run C (Gemini) fell between: three hypotheses, moderate revision depth, symmetric belief updating ($r = 0.00$). These behavioral profiles—including tool-use signatures that distinguish shell-heavy from search-heavy exploration—are invisible to output-only evaluation; they are precisely what the structured hypothesis ledger exists to capture. The martingale estimates ($n \leq 7$) are too few for reliable conclusions at this scale, but they are directionally consistent with the revision profiles: Run B’s $r = -0.43$ reflects predominantly downward confidence revisions, consistent with a run where updates tracked contradictory evidence (e.g., SIGLEC1’s non-specificity across interferonopathies, a judgment our independent human expert reviewer later corroborated).

The revision trail reveals reasoning invisible to a refutation-rate summary. Run B’s 0% refutation rate would, taken alone, suggest the agent never challenged its hypotheses—a flag that would cause a single-metric evaluation to dismiss the run as uncritical. The agent’s own evidence and confidence-revision trail, preserved in full by the hypothesis ledger, tells a different story: it explicitly acknowledged SIGLEC1’s cross-reactivity with COVID-19 interferonopathies, flagged absent GWAS support for GZMH, and distinguished diagnostic from

therapeutic utility for each candidate. Adversarial, self-critical reasoning was present throughout the confidence revisions; it just never surfaced as a formal status change, which is why the refutation count alone misses it. None of this requires a scoring layer to find—it is directly readable from the ledger itself, the structured trail OPERA keeps rather than discards.

The same trail also shows a case where that critical reasoning did not carry through. For GZMH, the agent *increased* its expressed confidence from 0.68 to 0.70 immediately after noting four explicit evidence gaps. Whether anything downstream caught this turned out to depend on which judge backend was asked. Claude Haiku flagged it directly: the agent’s justification, a “*strong 2nd modality*” from longitudinal clonal dynamics, cited no new GWAS, substrate, or mechanistic evidence, and Claude Haiku judged the recovery unsupported. Gemini Flash’s note on the same criterion instead read the agent’s own reasoning as reinforcing the evidence. The falsification observer, reading only the evidence independently, estimated $\hat{C} = 0.78$ for GZMH—*higher* than the agent’s own 0.70—offering no support for treating the raise as over-confidence either. Of three independent signals available for this one event, only one treated it as a problem. Independent human review later ranked GZMH lowest overall among Run B’s three candidates, consistent with the gaps the agent itself had flagged.

Judge backends differ in leniency but not in self-preference. Because both the evaluated agents and the LLM judges are drawn from the same two model families (Claude, Gemini), self-preference—a judge favoring output from its own family—is a standing risk of any LLM-as-judge design. The judge prompt carries no model or harness identifier, only the hypothesis, evidence trail, and agent rationale. This rules out *explicit* identity-based self-preference by construction; it does not test *implicit* stylistic self-recognition, since judges have been shown to recognize and favor their own outputs from writing style alone, without being told the source. Whether self-preference operates through an explicit identifier or implicit style recognition, it should leave the same signature in the scores themselves: same-family judge-agent pairs averaging higher than cross-family pairs. We tested for that signature directly by pooling every numeric criterion score across the three pilot runs and splitting by whether the judge backend shares a model family with the run’s agent. Same-family scores averaged *lower* than cross-family scores (3.23 vs. 3.51, $n = 39$ and 35 scale-rated criteria)—the opposite of what self-preference would predict. A more consistent pattern emerged instead: the Gemini Flash judge scored higher than the Claude Haiku judge in all three runs, including on Claude-generated work (Runs A, B) and relative to Claude Haiku judging Gemini’s own work (Run C)—a general leniency asymmetry between judge backends rather than a preference tied to output source, though implicit style recognition cannot be fully excluded with only three runs to disentangle it from calibration. We independently re-ran this analysis once more, rescoring all three runs with both backends from scratch: the same-family/cross-family direction held (3.30 vs. 3.45, $n = 40$ each) and Gemini Flash again scored higher than Claude Haiku in all three runs, by a wider margin in two of them. As with the martingale score, this check is too small ($n \leq 15$ per cell) to be conclusive, but it is a judge-bias question invisible to single-backend evaluation and one OPERA’s multi-backend design makes directly checkable.

A human-scored run reveals a judge-backend calibration gap. The judge-bias check above compares backends only to each other; we additionally obtained a fully human-scored review of Run B from a domain expert (an SLE researcher), scoring blind to both judges’ outputs per OPERA’s review protocol (§3). Against this human anchor, the two backends diverge sharply (Table 2). Claude Haiku’s per-criterion scores track the human reviewer closely (mean absolute difference 0.42 on a five-point scale, $n = 12$ rated criteria, errors balanced in both di-

Table 2: Per-hypothesis summed rubric scores for Run B: human reviewer vs. both LLM judge backends (4 per-hypothesis criteria, max 20).

Hypothesis	Human	Claude Haiku	Gemini Flash
IFI44L	14	14	17
SIGLEC1	13	10	17
GZMH	10	10	16

rections), and its ranking places IFI44L on top as she did, though it ties SIGLEC1 and GZMH at 10 each where she distinguished them (13 vs. 10). Gemini Flash diverges substantially, with a mean absolute difference of 1.08—more than double Claude’s—and scores higher than the human reviewer in 11 of 12 comparisons; its own ranking barely differentiates the three hypotheses, clustering at 16–17. This is the same compression toward uniformly high, undifferentiated assessments observed above for Gemini *as an agent* in Run C (flat martingale, confidence uniformly high regardless of evidence). One human-scored run cannot establish which backend is better calibrated in general, and the narrow 3-point gap underlying Claude Haiku’s ranking match is itself fragile—independent re-runs of the same judge on this hypothesis set did not consistently reproduce it, with SIGLEC1 and GZMH swapping relative rank across attempts. The much larger, consistently one-sided gap between backends (0.42 vs. 1.08 mean absolute difference) is not the kind of thing reshuffled by rescoring noise, though, and remains at least suggestive that the leniency asymmetry above is not merely an artifact of comparing two LLMs to each other: one of them tracks an independent domain expert markedly better than the other.

Cross-run convergence on IFI44L. IFI44L emerged as the top-ranked hypothesis in all three independent runs—Runs A and B (Claude) and Run C (Gemini)—despite each using different tool strategies and different starting points in their searches. Table 3’s **Recurs.** column, extracted directly from the structured hypothesis ledger, makes this convergence easy to verify: IFI44L recurs in all three runs, SIGLEC1 in two. Convergence alone, however, is not evidence of independent discovery. Our domain-expert reviewer rated IFI44L the strongest of Run B’s three candidates and the only one she judged deserved to survive unrefuted, but also flagged it as an established, non-novel marker. The runs’ own evidence agrees: Run A cites it as “a core component of multiple validated IFN signature scores used clinically for SLE diagnosis,” and Run C calls it a “well-established diagnostic biomarker.” All three runs, in other words, converged on the same well-characterized prior finding rather than independently discovering something new. What did differ across runs was how much evidence each assembled beyond that shared baseline: Run B alone contributed a causal risk SNP (rs273259), a trans-eQTL network link, knockout-mouse evidence of a disease-modifying effect, and a mechanistic IFN- α /c-Jun/AP-1 axis (agent confidence 0.88; falsification-observer confidence 0.89), while Run C’s supporting evidence stayed at the level of the shared prior literature. Cross-run agreement is thus a meaningful signal for candidate identification, but not, on its own, for depth of investigation—a distinction the aggregate convergence count alone does not surface.

5 Discussion and Limitations

Human expert availability. The human review step requires a domain scientist, which limits throughput. We considered a two-stage design in which the LLM judges score first and the

Table 3: Per-hypothesis summary across all three pilot runs. Rev.: confidence-updating revisions. Evid.: recorded evidence items. b_T : final agent-expressed confidence (R=refuted). $\hat{C}$: independent observer confidence derived from evidence trail only. Recurs.: runs in which the gene was independently nominated.

Run	Gene	Rev.	Evid.	b_T	$\hat{C}$	Recurs.
A	IFI44L	0	3	0.92	0.89	3/3
	SIGLEC1	1	3	0.75	0.72	2/3
	TNFSF13B	0	3	0.80	0.92	1/3
	LILRA4	0	2	0.55	0.72	1/3
	IKZF2	1	2	R	—	—
B	IFI44L	2	5	0.88	0.89	3/3
	SIGLEC1	2	5	0.80	0.87	2/3
	GZMH	3	4	0.70	0.78	1/3
C	IFI44L	1	3	0.80	0.72	3/3
	ISG15	1	3	0.80	0.62	1/3
	MX1	1	3	0.80	0.72	1/3

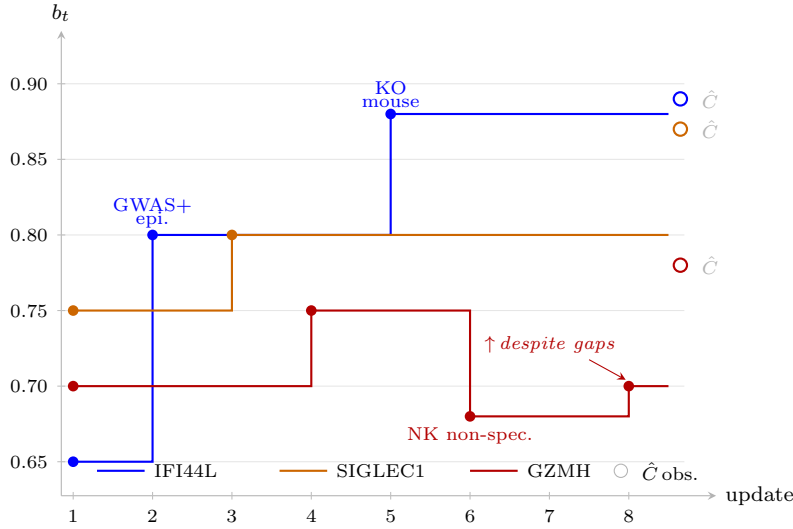


Figure 2: Belief trajectories for Run B (Claude Sonnet 4.6). Each step reflects a confidence-updating ledger revision; events are numbered in chronological order. All three hypotheses begin within seconds and diverge as evidence accumulates. Hollow circles ($\hat{C}$) show the independent observer’s confidence computed from the evidence trail alone—all three sit above the agent’s final b_T , confirming appropriately conservative self-reporting. The GZMH trajectory (red) rises at event 8 despite the agent explicitly cataloguing four evidence gaps at event 6, a pattern detectable only in the revision trail.

reviewer adjudicates only the criteria where backends disagree, triaging the expert’s attention to contested cases. We chose full blind review instead: the reviewer scores every criterion independently, with LLM scores revealed only after submission, to avoid anchoring the human’s

judgment on any judge’s output. This trades reviewer throughput for an uncontaminated calibration point—the full per-criterion comparison reported above would not be possible from a review restricted to contested cases alone. A triage-based interface remains a reasonable choice when throughput matters more than an independent baseline.

A domain gap invisible to both the agent and its LLM judges. The human review flagged a gap that neither Run B’s own hypotheses nor either judge backend’s rationale caught: none of its three candidates were B-cell- or neutrophil-specific, despite B cells being an established SLE driver. Recognizing an absent candidate class takes clinical and immunological knowledge that neither the agent generating hypotheses nor the LLM judges scoring them has. This blind spot sits at both ends of the pipeline, not in scoring alone. The likely cause is a task-level one. Across all three runs, scRNA-seq analysis queried only the seed Perez et al. 2022 PBMC cohort, and neither “neutrophil” nor “granulocyte” appears anywhere in any run’s hypothesis ledger. Neither term was ever raised, let alone considered and ruled out. The task specification gives this one dataset a full access recipe and names no alternative; despite having unconstrained tool access in principle (§4), no run sought a different primary data source (the closest exception, Run B’s single Human Protein Atlas lookup, checked one gene’s tissue specificity rather than searching for cell-type coverage more broadly). This points to a task-design constraint rather than a demonstrated agent-capability limitation, though with only three runs we cannot rule out the latter. Human review surfaced not just a gap in the results but a plausible mechanism behind it. Automated scoring could not have raised this on its own, since it sees what the agent proposed but not what data the agent never sought.

Rubric design lessons from the first human review. Piloting the review process with a domain expert (an SLE researcher, scoring Run B) surfaced lessons for rubric-driven evaluation that are not specific to OPERA’s biomedical setting. The most consequential lesson was that rubric criteria gave a scale and a description of what to weigh, but no anchors for what evidence earns each score. As a result, reviewers, whether human or LLM, had no shared reference point to calibrate against. Behaviorally-anchored criteria, with example evidence illustrating a low versus a high score, together with a task-level floor scenario showing a minimal-effort baseline run, would let human and LLM judges converge on the same scale rather than each inventing their own. Two further risks apply to rubric-driven evaluation generally, regardless of who is scoring. The first is *evidence locality*. When a rubric poses several separate questions against one shared evidence trail, a reviewer, human or LLM, can lose track of which evidence item supports which criterion. A reviewer might, for example, credit *clinical relevance* with evidence that actually supports *evidence strength*, simply because both draw on the same trail read once at the outset. The second is *construct conflation*, when guidance bundles several separable questions under one criterion. Our novelty criterion did exactly this, asking at once about prior literature grounding, evidence integration, and clinical impact. The result is a single score whose meaning is hard to interpret afterward. Because OPERA’s rubric is injected verbatim into both the LLM judge and the human form (§3), addressing any of these strengthens both evaluation layers at once.

Ground truth will eventually exist. For some tasks (e.g., proposed biomarkers that are later validated in independent studies), retrospective ground truth becomes available. OPERA’s rubric-based scoring can be compared against such retrospective labels as a long-term calibration signal.

Small-sample process metrics. The martingale score and the judge-backend calibration check in §4 both pool few data points per run ($n \leq 7$ revision pairs and $n \leq 15$ per judge/family cell, respectively). At this scale both are descriptive signals, not statistically powered tests; we report them because they are directionally consistent with independently observed behavior (revision depth, refutation rate) and because no single-layer metric would surface them at all, but neither should be read as a confirmed effect until validated on more runs.

Harness coverage and generalization. Because OPERA evaluates harnesses rather than models, its findings do not generalize directly to model capability: a result obtained with one scaffolding strategy may not hold under a different prompting approach, tool configuration, or agent framework. Currently, Claude Code and Gemini CLI serve as reference harnesses; the MCP interface allows additional agent frameworks to be integrated. Systematic comparison across harnesses for the same underlying model is a natural next step and would help disentangle model contribution from scaffolding contribution in observed performance.

Applicability and future extensions. OPERA is best suited to discovery tasks that admit a structured hypothesis representation—a candidate, a claim, accumulating evidence, and a revisable confidence—rather than tasks whose output is a single artifact evaluated holistically (e.g., code generation, writing). Within that class, we expect the framework to generalize readily to other biomedical discovery problems (drug repurposing, pathway analysis, variant interpretation) and, with the ledger schema adapted to the domain’s evidence types, to non-biomedical open-world discovery more broadly. Two extensions we leave to future work: scaling the pilot evaluations across additional biomedical sub-domains (e.g., structural biology alongside translational genomics) to establish generalizability at greater breadth, and a sensitivity analysis of how minor prompt variations in the Evaluator persona affect multi-backend judge consensus. Both require substantially more pilot runs than the present study’s scale supports. The evaluation itself has a narrower scope than the framework. All pilot runs use one task domain, and consensus scoring rests on two judge-backend families rather than three or more. Nor have we ablated the reviewer pipeline’s own design choices. Future work will test how two-pass scoring, an independent falsification observer, and multi-backend contested-criterion flagging each compare to a simpler single-backend, single-pass judge.

Preliminary results. This paper presents the OPERA framework and pilot task instantiations. Human expert scoring of one pilot run is reported in §4; extending it to the remaining runs is ongoing. We release the codebase and task specifications to enable the community to reproduce and extend our evaluations.

6 Conclusion

OPERA provides a principled framework for evaluating LLM agents on open-world biomedical discovery tasks. By separating task specification, agent execution, and multi-modal review into three well-defined personas with structured artifact handoffs, and by introducing process-level metrics (martingale score, falsification observer, two-pass LLM judging), OPERA enables evaluation of scientific reasoning quality in the absence of ground truth. We argue this evaluation paradigm is necessary as AI systems are deployed on frontier scientific problems—and we release OPERA as an open framework for the community to extend.

Code and task specifications: <https://github.com/LanGuo/opera>

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A SLE Biomarker Discovery Task Specification

The declarative specification backing the example introduced in §1 and instantiated in §4 is listed below in full (em dashes normalized to -- for the monospace listing font). The rubric block is the source of truth for scoring: the same guidance text is injected into both the LLM judge’s prompt and the human reviewer’s form.

```
task_id: sle_biomarker_discovery
goal: >
  Identify the top 3 cell-type-specific candidate biomarkers or therapeutic targets
  for systemic lupus erythematosus (SLE / lupus), supported by multi-modal evidence.
  Starting data: Perez et al. 2022 SLE PBMC cohort, accessible via the cellxgene_census
  Python package (dataset ID: perez_2022, approximately 1.2 million cells, 162 donors --
  SLE patients and healthy controls). The agent may discover and use additional datasets.
termination_criteria:
  min_hypotheses_with_multimodal_evidence: 3
  max_tool_calls: 200
evaluator:
  model: claude-opus-4-7
  max_turns: 250
rubric:
  - id: novelty
    label: "Scientific Novelty"
    scope: per_hypothesis
    scale: "1-5"
    guidance: >
      Is this gene/cell-type combination a well-known SLE association or genuinely novel?
      Does multi-modal evidence (DEG, GWAS, protein interaction, literature) strengthen
      the claim beyond what is already established?
  - id: experimental_design
    label: "Experimental Design Quality"
    scope: per_hypothesis
    scale: "1-5"
    guidance: >
      Feasibility of proposed validation (realistic cell numbers, reagents, assays),
      specificity to the hypothesis, statistical power (cohort size and controls),
      and translatability to clinical use.
  - id: evidence_strength
    label: "Evidence Strength"
    scope: per_hypothesis
    scale: "1-5"
    guidance: >
      Quality and independence of evidence modalities. Are at least 2 independent
      sources (scRNA-seq DEG, GWAS, protein interaction, literature, drug database)
      supporting the hypothesis with quantified effect sizes?
  - id: clinical_relevance
    label: "Clinical/Translational Relevance"
    scope: per_hypothesis
    scale: "1-5"
    guidance: >
      Potential clinical impact if validated -- relevant to SLE diagnosis, disease
      monitoring, or treatment. Is the target druggable or measurable in clinical
      samples?
  - id: reasoning_quality
    label: "Reasoning Quality"
    scope: overall
    scale: "1-5"
    guidance: >
      Did the agent update beliefs appropriately on evidence, handle contradictory
      findings, and seek diverse modalities rather than cherry-picking confirming
      evidence? Were refutations justified?
  - id: overall_quality
    label: "Overall Scientific Quality"
    scope: overall
    scale: "1-5"
    guidance: >
      Overall scientific rigor and quality of the discovery run -- appropriate
      statistical methods, multi-modal evidence integration, and depth of exploration.
  - id: wet_lab_worthy
    label: "Worth Pursuing in Wet Lab"
```

```

scope: overall
scale: "yes/no/partial"
guidance: >
  Would the top hypotheses be worth pursuing in a wet lab validation study?
  Consider feasibility, novelty, and clinical relevance together.

```

B Protein Synergy Discovery Task Specification

The declarative specification for the second task introduced in §4 is listed below in full (em dashes normalized to -- for the monospace listing font). This task has not yet been run; it is included to demonstrate that the specification format transfers to a structurally different discovery problem, where the agent must both find its own dataset and explain a quantitative pattern within it, rather than evaluate candidates against an externally provided reference dataset.

```

task_id: protein_synergy_discovery
goal: >
  Investigate the synergistic (non-additive) effects of double mutations in the
  protein GB1 (or similar fitness landscape datasets). Determine if synergy
  is primarily driven by proximity to the active site or by physical distance
  between the two mutation sites. The agent must discover and use a public
  fitness landscape dataset (e.g., from ToolUniverse, GitHub, or academic repositories).
termination_criteria:
  min_variations_explored: 3
  journal_entries_count: 5
  reproducible_validation_script: true
  max_tool_calls: 150
evaluator:
  model: claude-opus-4-7
  max_turns: 200
rubric:
- id: hypothesis_novelty
  label: "Hypothesis Novelty"
  scope: per_hypothesis
  scale: "1-5"
  guidance: >
    Is the proposed synergy driver (proximity to active site vs. physical distance)
    a well-known result or a finding that adds new understanding from this specific
    dataset analysis?
- id: evidence_strength
  label: "Evidence Strength"
  scope: per_hypothesis
  scale: "1-5"
  guidance: >
    Quality of evidence for the synergy hypothesis -- appropriate statistical methods,
    adequate sample size from the fitness landscape, and reproducibility of the
    key finding.
- id: visualization_correctness
  label: "Visualization Correctness"
  scope: overall
  scale: "1-5"
  guidance: >
    Check if activity heatmaps use sequential color scales and divergent scales
    only for relative changes (delta, synergy scores). Using a divergent scale
    on raw fitness [0,1] data is a scoring error.
- id: interview_quality
  label: "Interview / Context-Loading Quality"
  scope: overall
  scale: "1-5"
  guidance: >
    Did the agent ask at least 2 clarifying questions about the data schema or
    biological active site before performing analysis? This tests whether the
    agent loads context before diving in.
- id: dataset_discovery
  label: "Dataset Discovery"
  scope: overall

```

```
scale: "yes/no/partial"
guidance: >
  Did the agent autonomously find and cite a relevant protein fitness landscape
  dataset (e.g., GB1, GFP, or similar deep mutational scanning data) without
  being told which one to use?
- id: overall_quality
label: "Overall Scientific Quality"
scope: overall
scale: "1-5"
guidance: >
  Scientific rigor of the analysis including reproducibility (standalone script),
  appropriate statistical methods, depth of exploration across at least 3
  distinct visualization approaches, and journal quality.
```